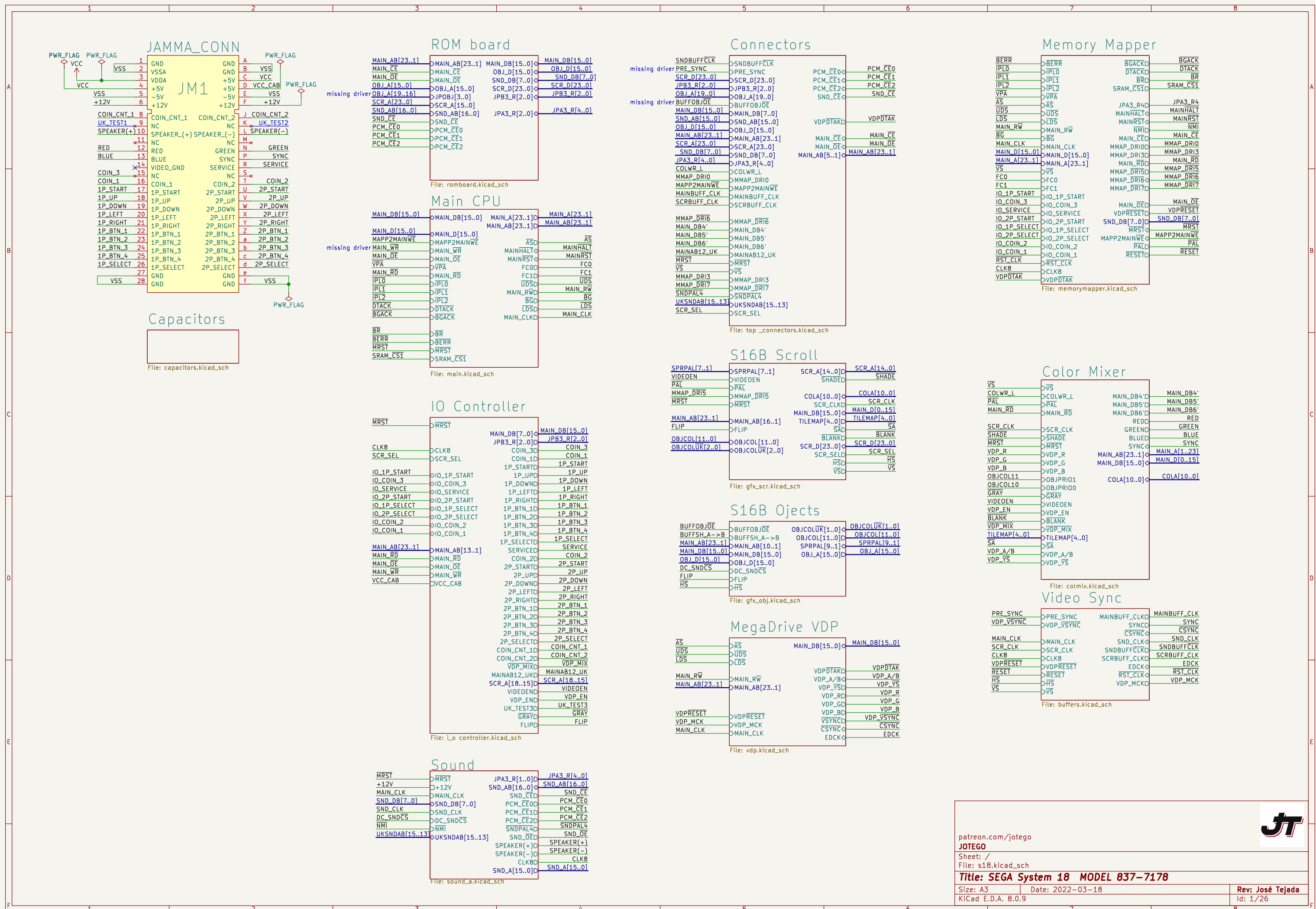
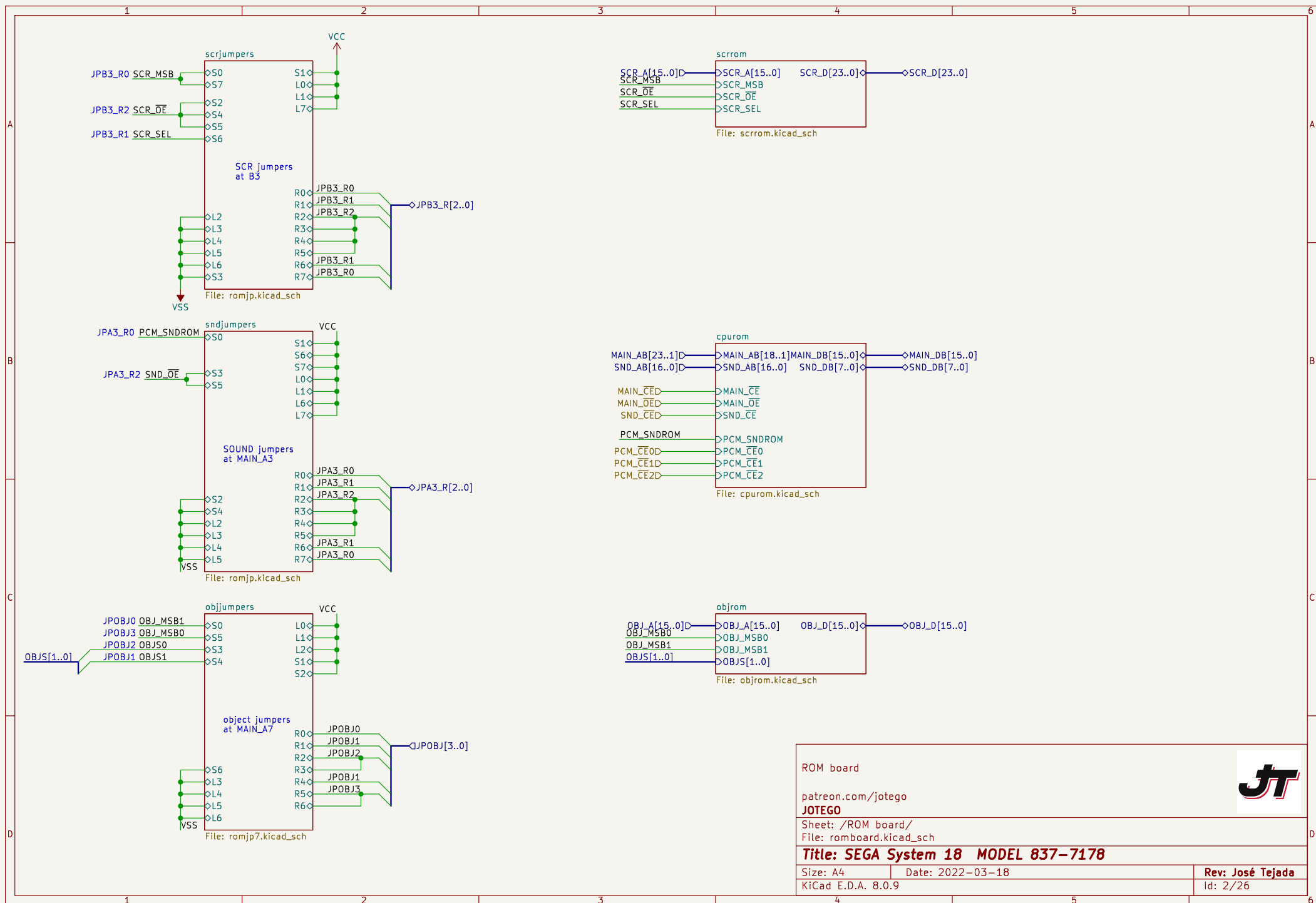






ROM board

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Sheet: /ROM board/

File: romboard.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4

Date: 2022-03-18

KiCad E.D.A. 8.0.9

Rev: José Tejada

Id: 2/26





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Sheet: /ROM board/scrjumpers/

File: romjp.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4

Date: 2022-03-18

Rev: José Tejada

KiCad E.D.A. 8.0.9

Id: 3/26





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Sheet: /ROM board/sndjumpers/

File: romjp.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4

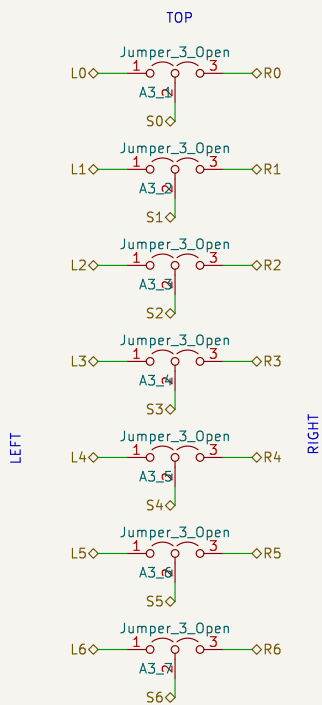
Date: 2022-03-18

Rev: José Tejada

KiCad E.D.A. 8.0.9

Id: 4/26





BOTTOM

Seven jumpers

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Sheet: /ROM board/objjumpers/

File: romjp7.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4

Date: 2022-03-18

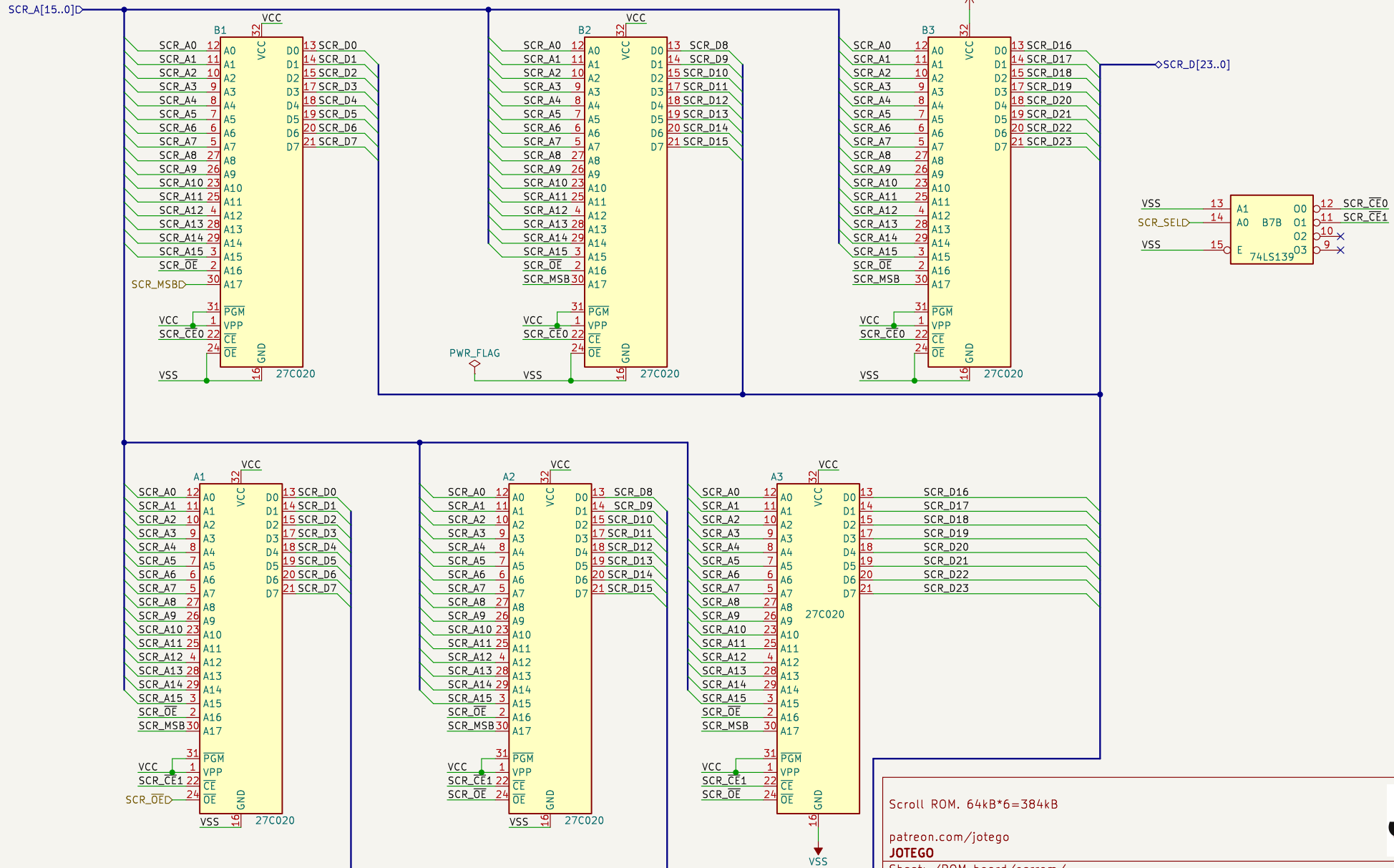
Rev: José Tejada

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Id: 5/26



SCROLL ROM



Scroll ROM. 64kB*6=384kB

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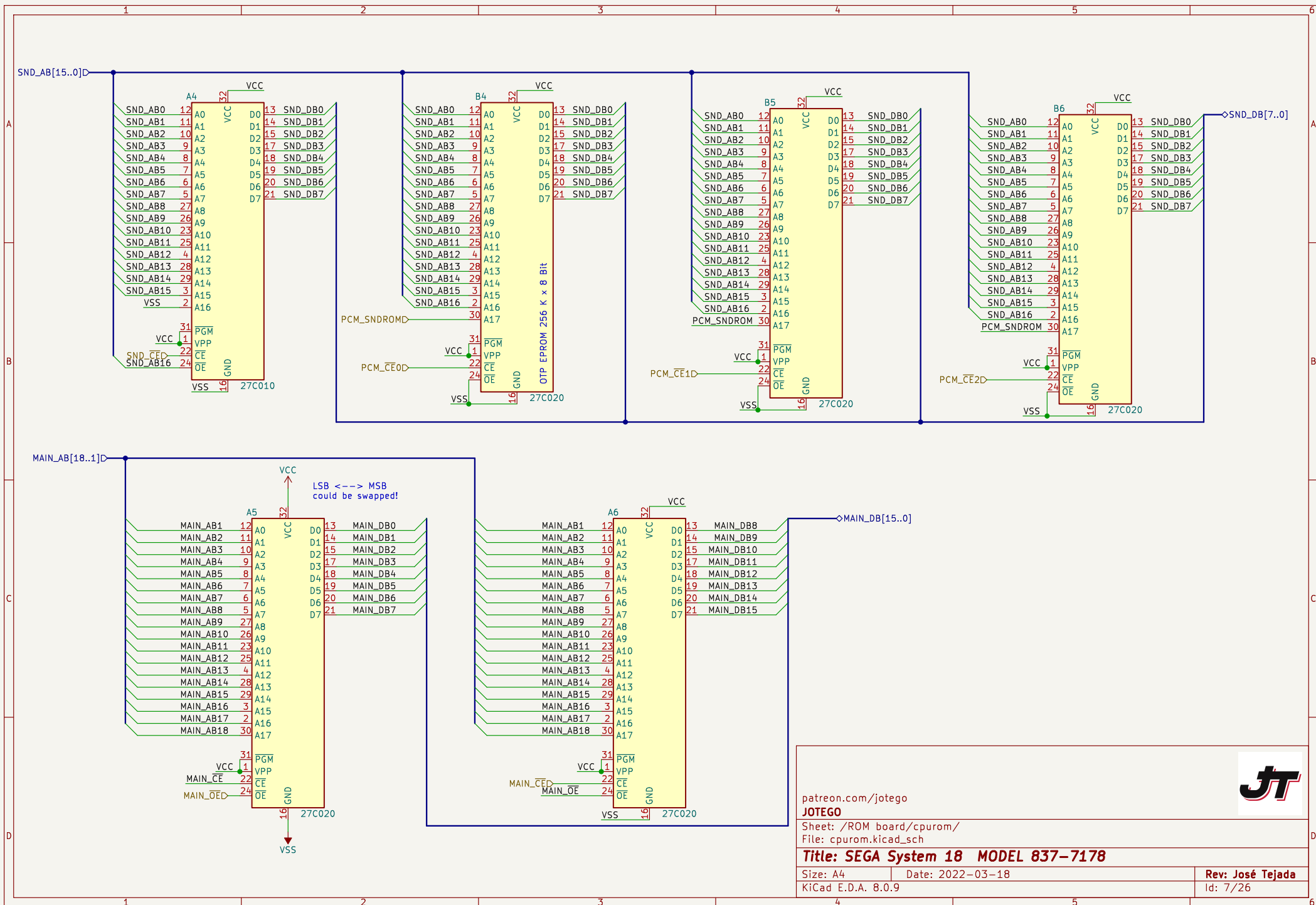
Sheet: /ROM board/scrrom/
File: scrrom.kicad_sch

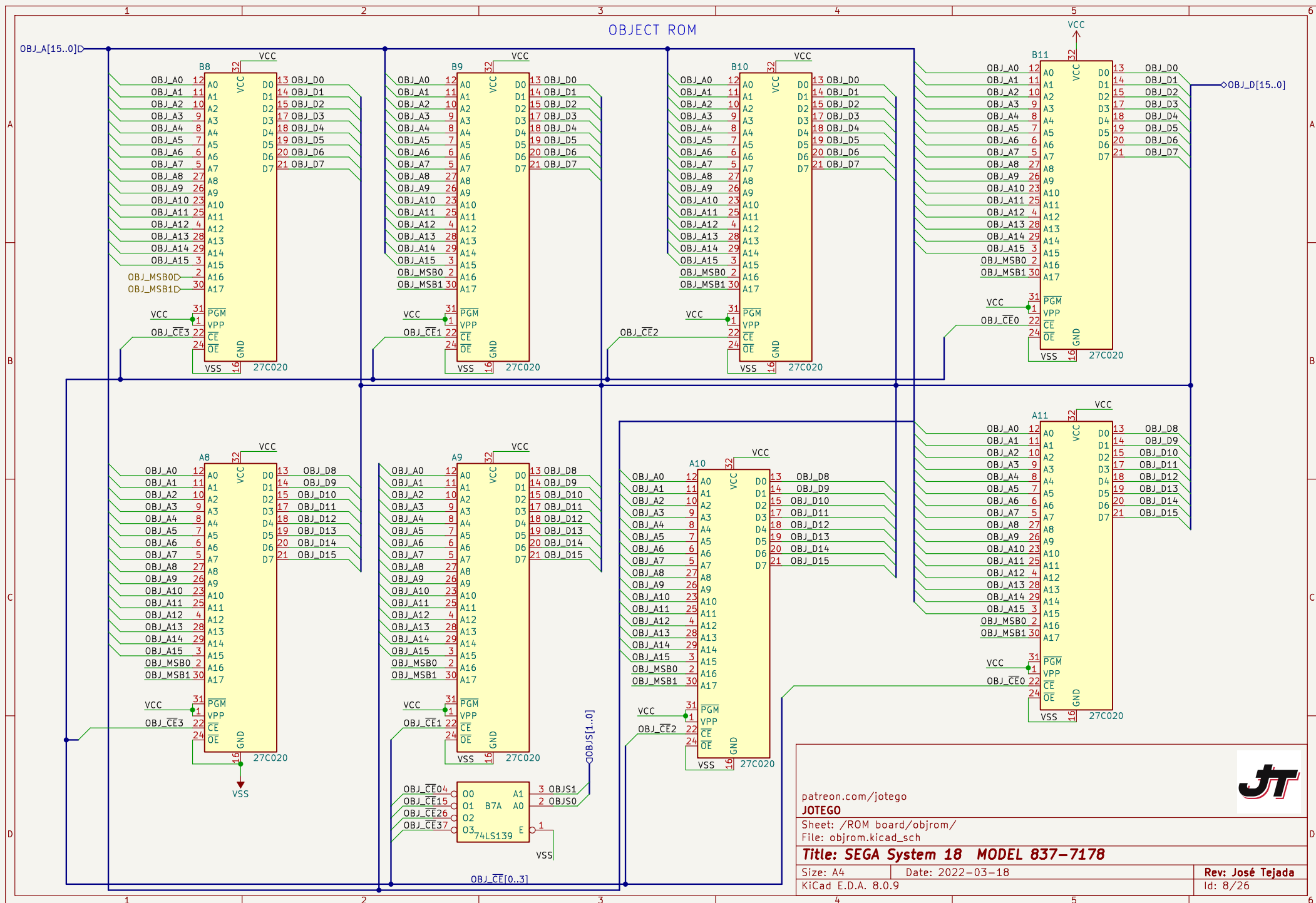
Title: SEGA System 18 MODEL 837-7178

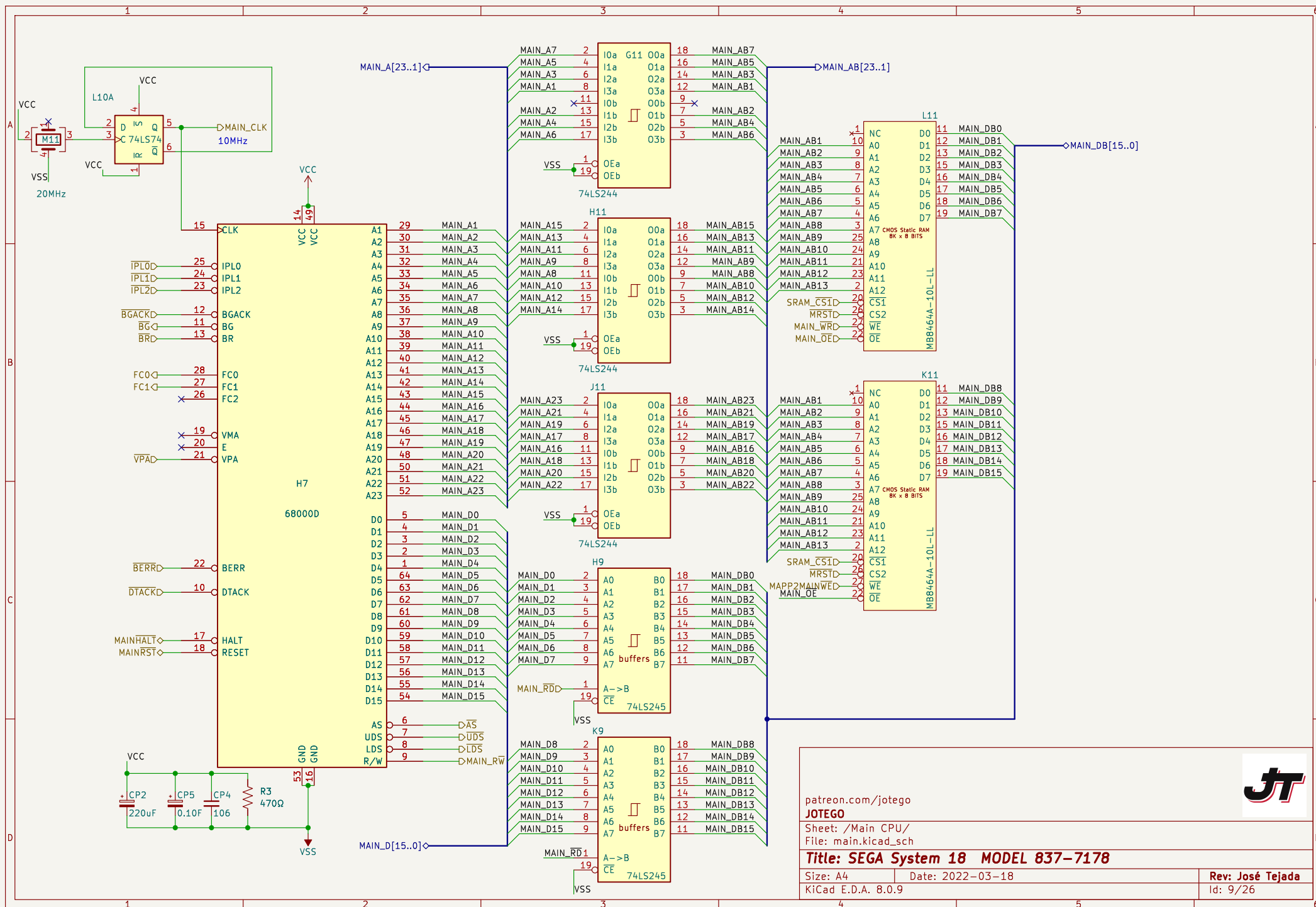
Size: A4 Date: 2022-03-18
KiCad E.D.A. 8.0.9

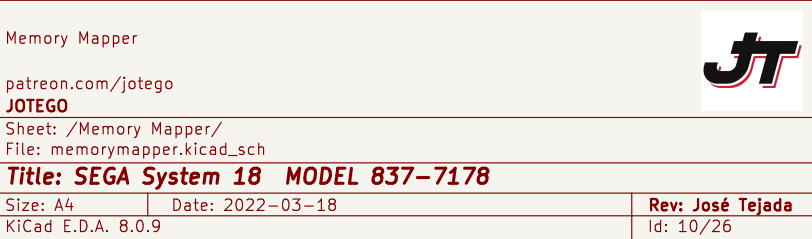
Rev: José Tejada
Id: 6/26

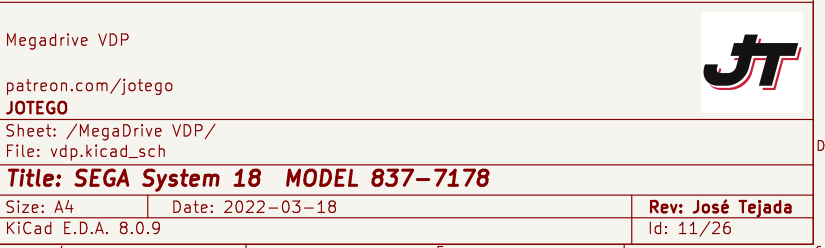






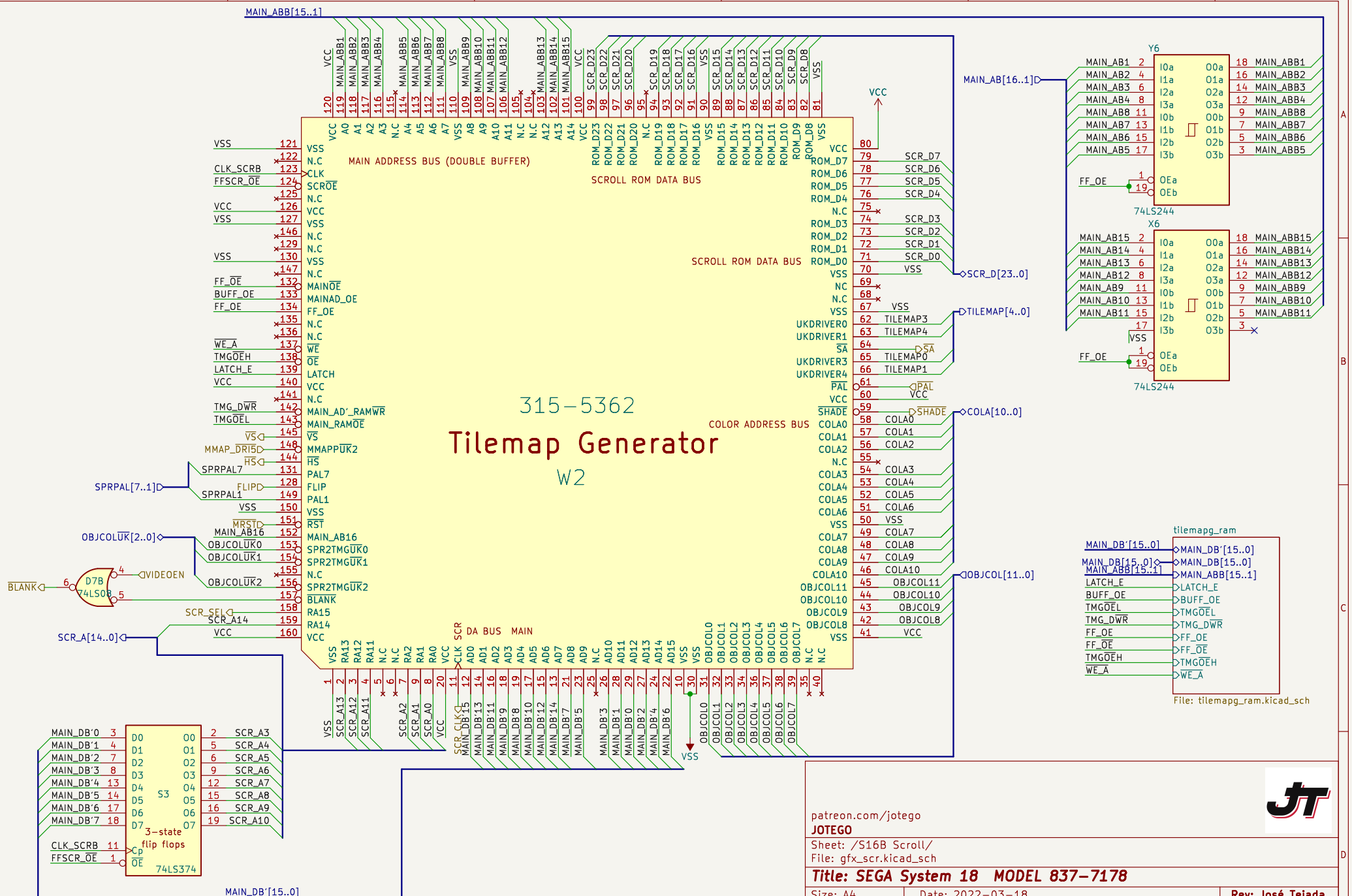






Tilemap Generator

315-5362
W2



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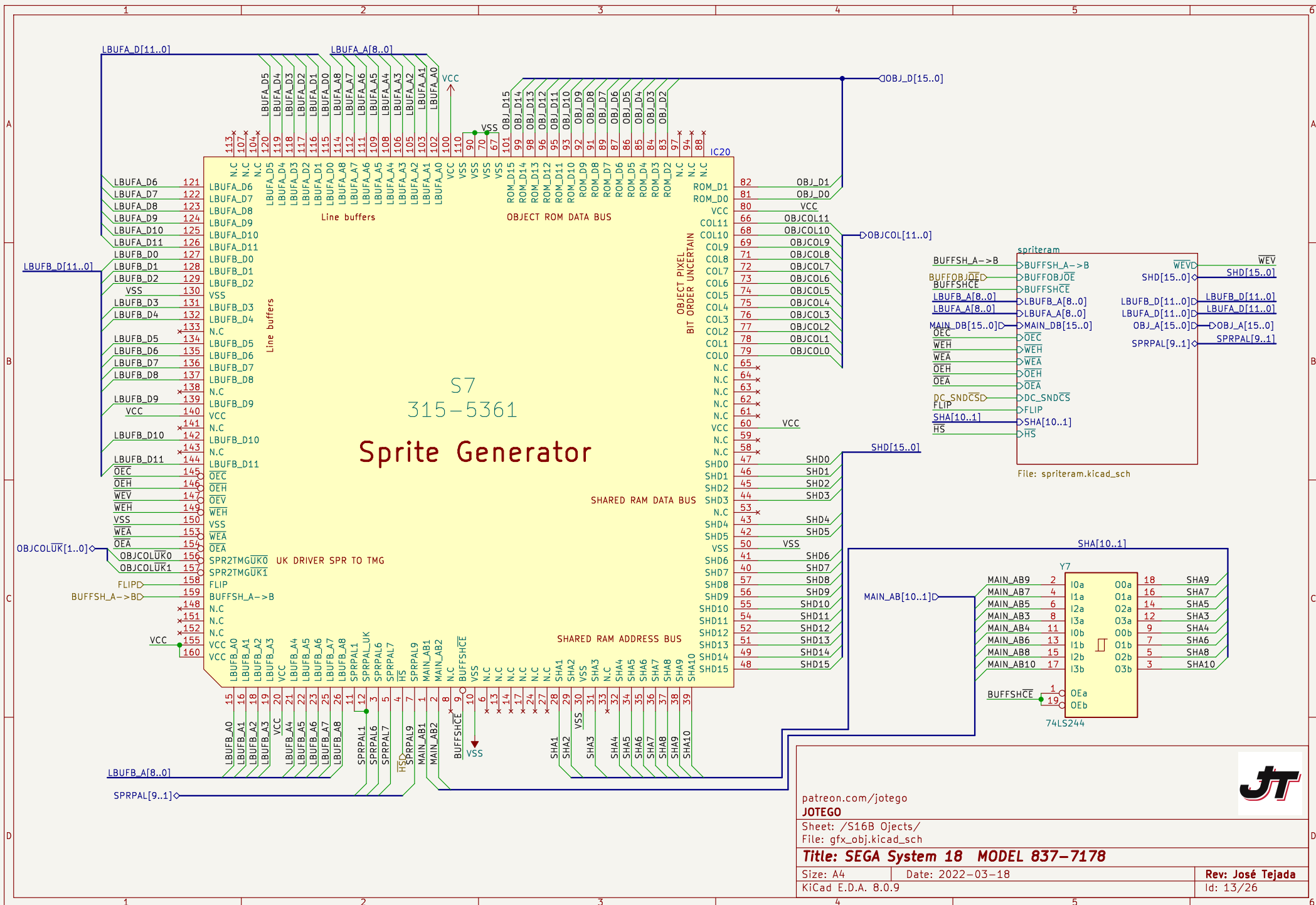
Sheet: /S16B Scroll/
File: gfx_scr.kicad_sch

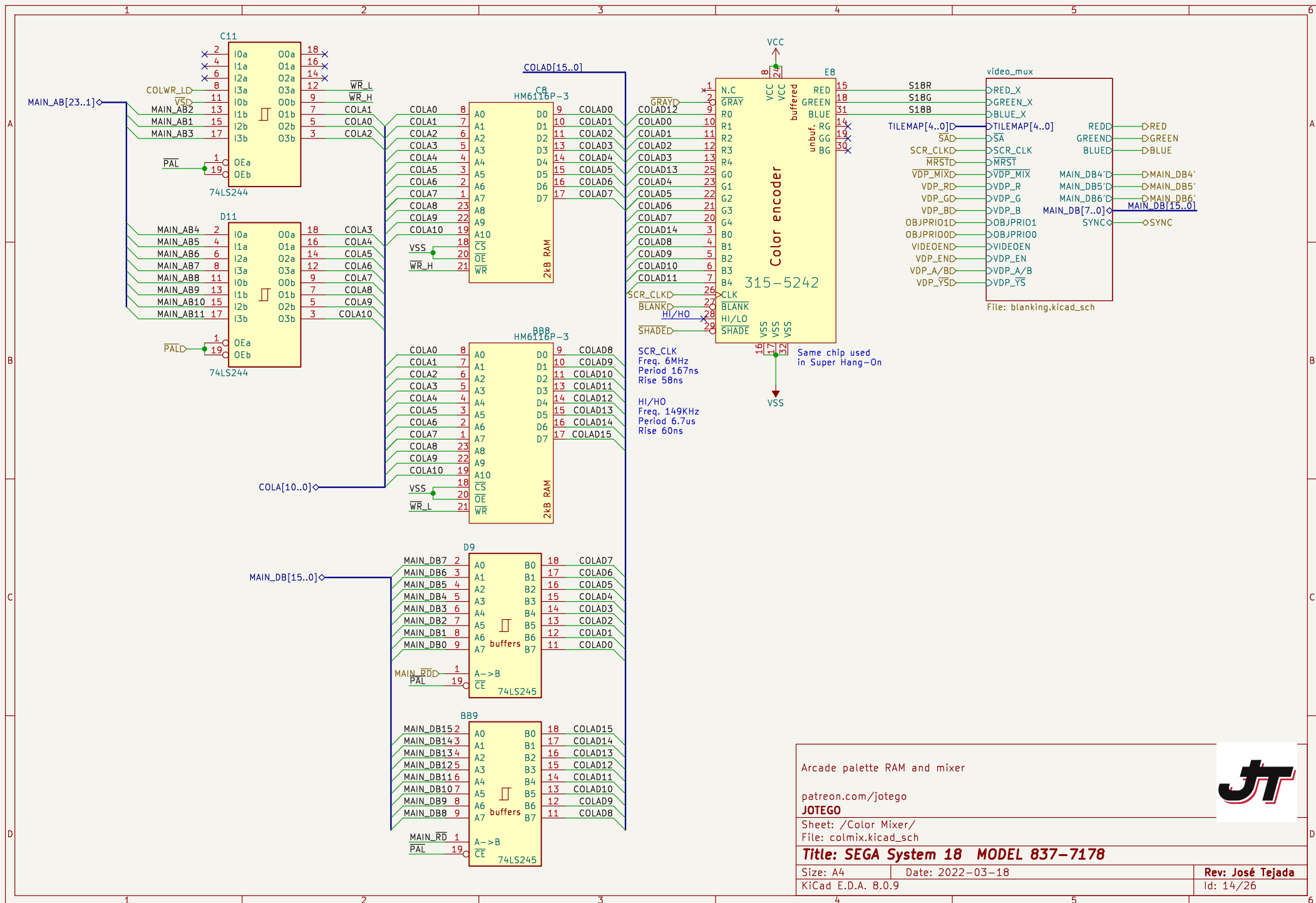
Title: SEGA System 18 MODEL 837-7178

Size: A4
Date: 2022-03-18
KiCad E.D.A. 8.0.9

Rev: José Tejada
Id: 12/26







Arcade palette RAM and mixer

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Sheet: /Color Mixer/

File: colmix.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4

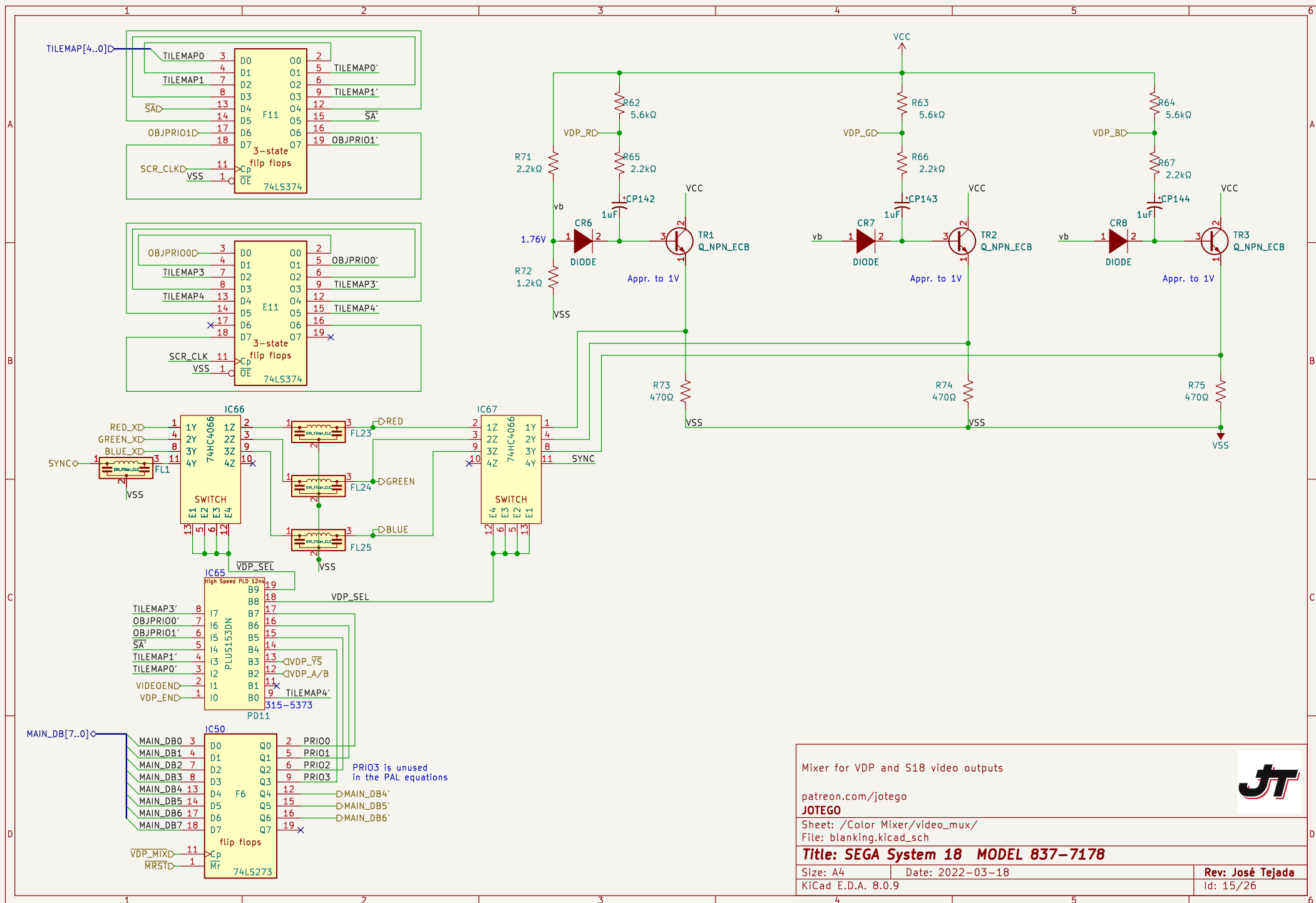
Date: 2022-03-18

KiCad E.D.A. 8.0.9

Rev: José Tejada

Id: 14/26





Mixer for VDP and S18 video outputs

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Sheet: /Color Mixer/video_mux/

File: blanking.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4 Date: 2022-03-18

KiCad E.D.A. 8.0.9

Rev: José Tejada

Id: 15/26





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Sheet: /Capacitors/
 File: capacitors.kicad_sch

Title: SEGA System 18 MODEL 837-7178

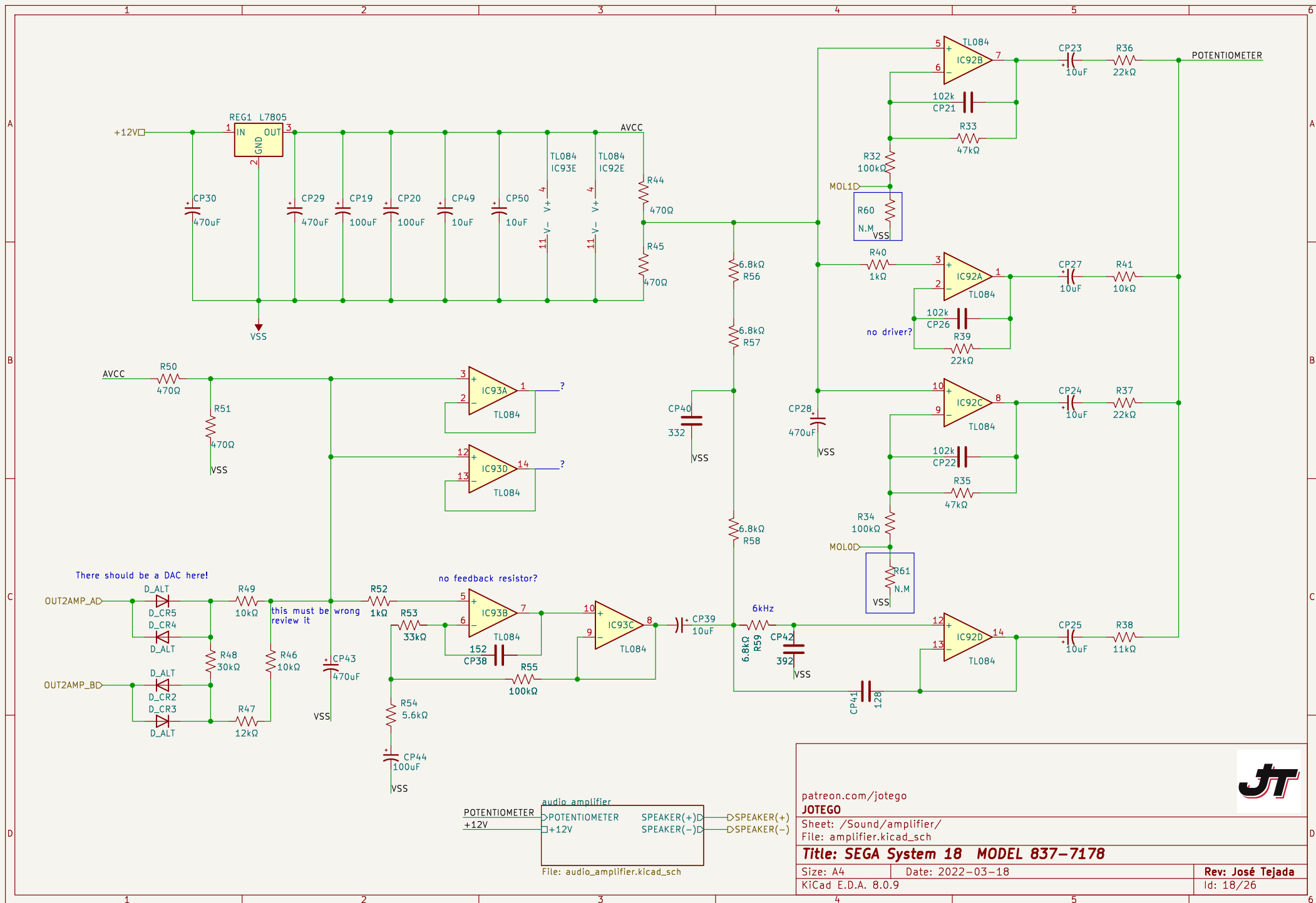
Size: A4 Date: 2022-03-18
 KiCad E.D.A. 8.0.9

Rev: José Tejada
 Id: 16/26

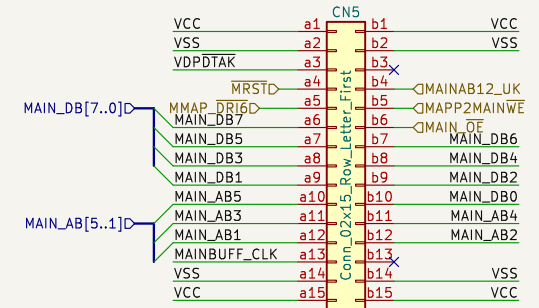
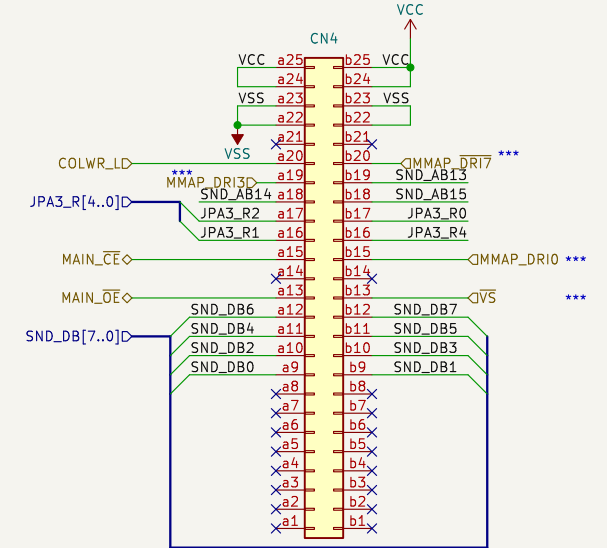
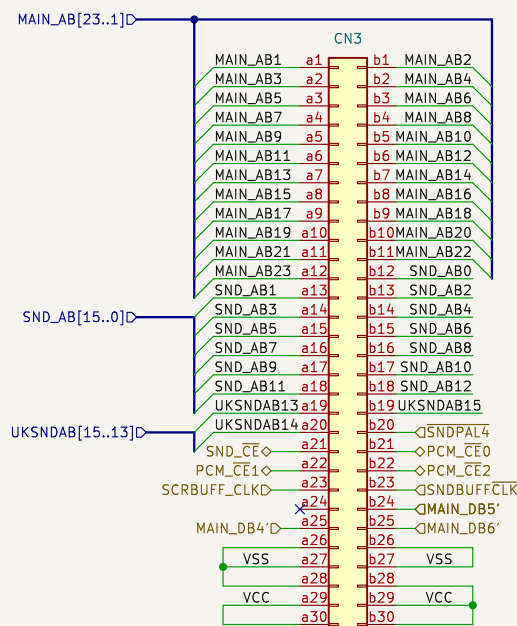
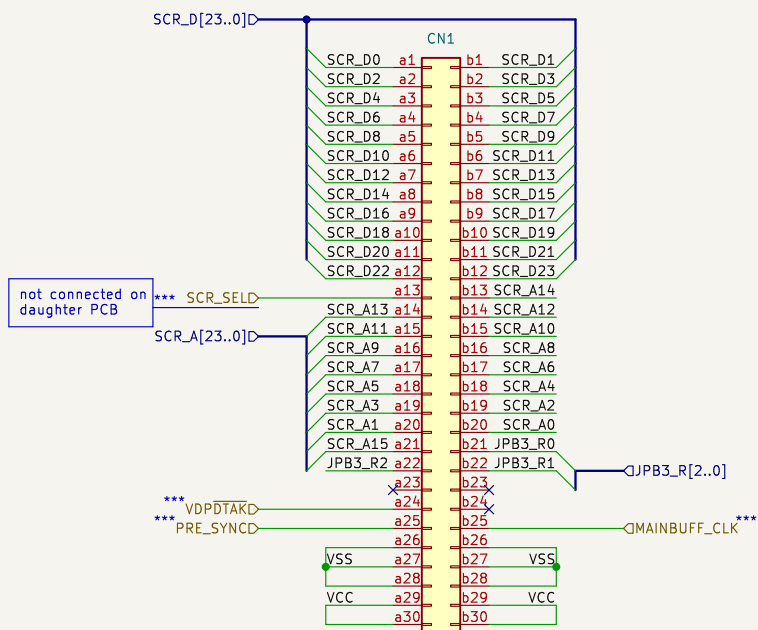




Rev: José Tejada
Id: 17/26



MOTHER AND DAUGHTER PCB CONNECTORS



Used as ADC inputs from gun in Laser Ghost on board: 837-7536 171-5993A

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Sheet: /Connectors/
File: top_connectors.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4
KiCad E.D.A. 8.0.9

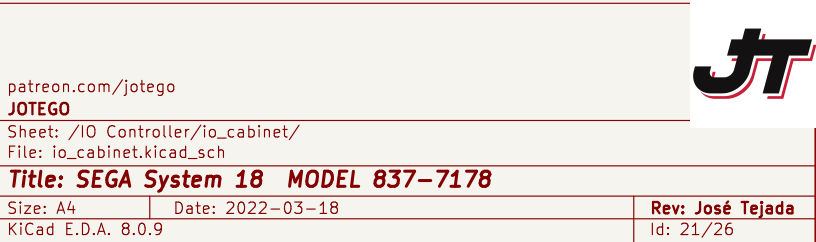
Date: 2022-03-18

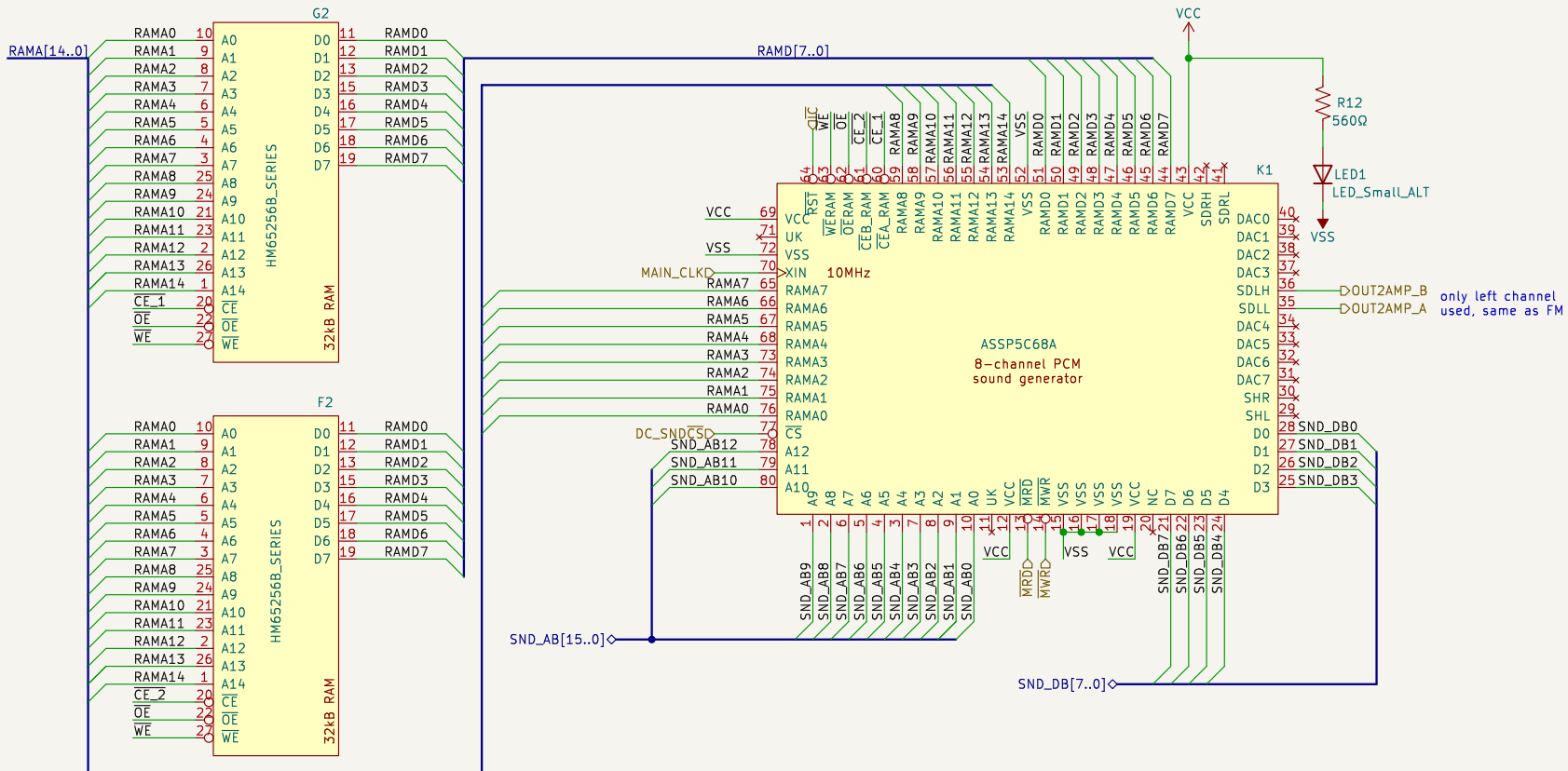
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Id: 19/26





Id: 20/26





PCM sound generator

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Sheet: /Sound/PCM/

File: comm_snd.kicad_sch

Title: SEGA System 18 MODEL 837-7178

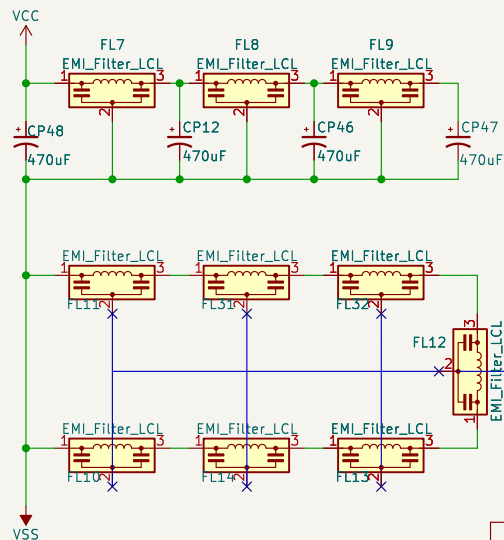
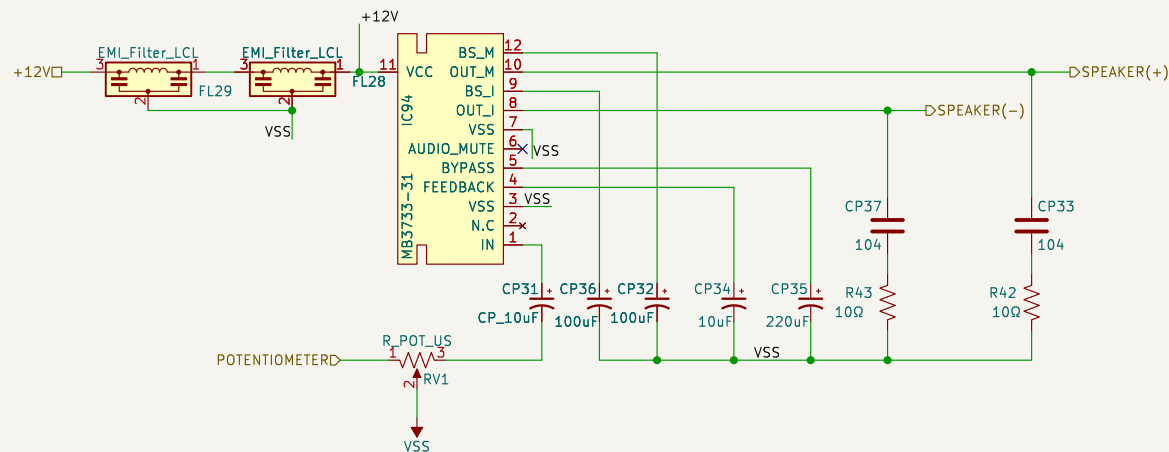
Size: A4 Date: 2022-03-18

KiCad E.D.A. 8.0.9

Rev: José Tejada

Id: 22/26





Leaded EMIFL
Three-terminal Capacitor type
DSS6NE52A222p
2200pF 80/-20%
100Vdc 6A -25 to +85°C

A kind of a track is observed between the two PCB layers, as if it were another layer, but there is no certainty about that. The track seems to be used to delimit or isolate the EMI (ElectroMagnetic Interference) filters. They may have wanted to isolate the noise between digital and analog devices.

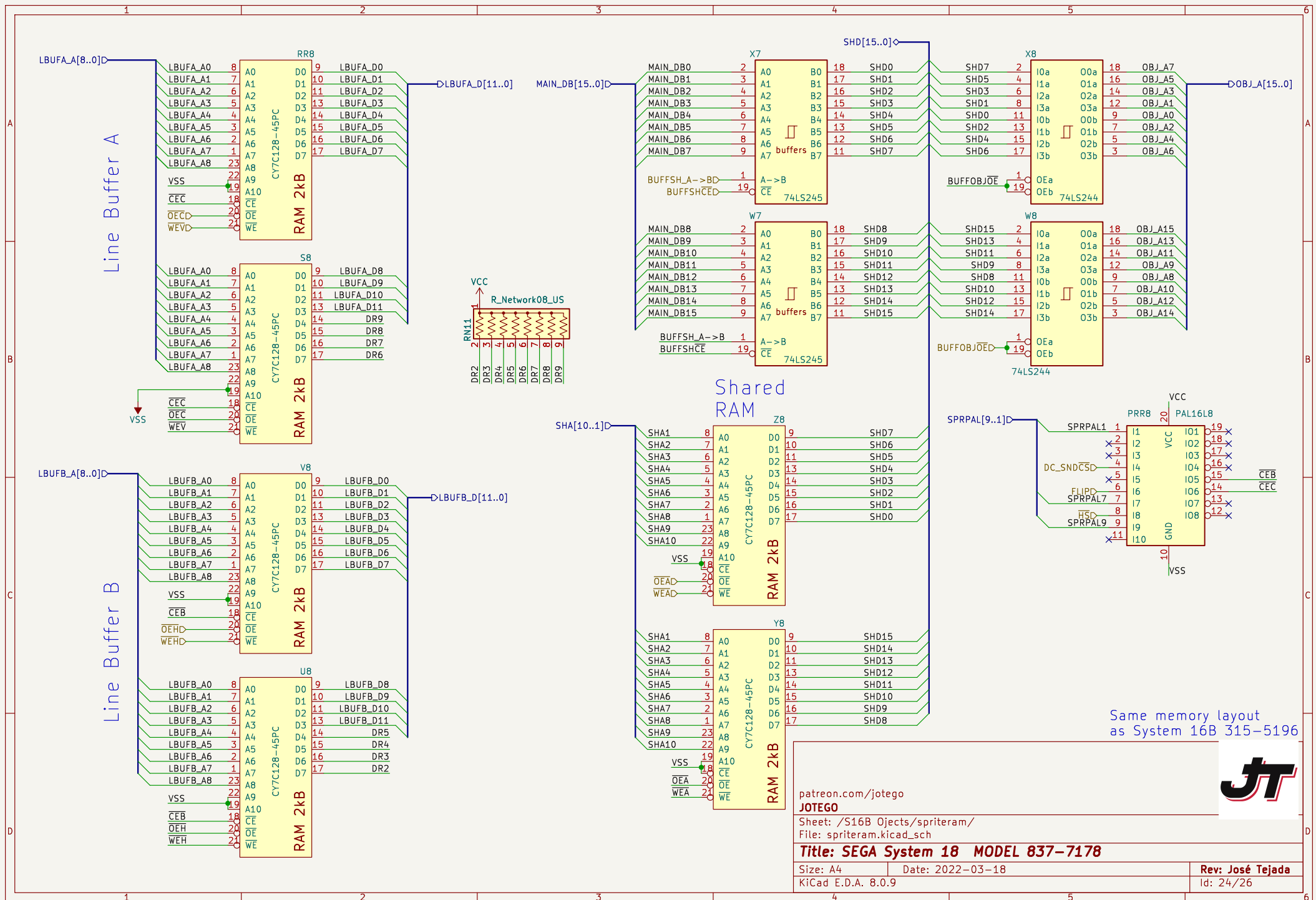
patreon.com/jotego
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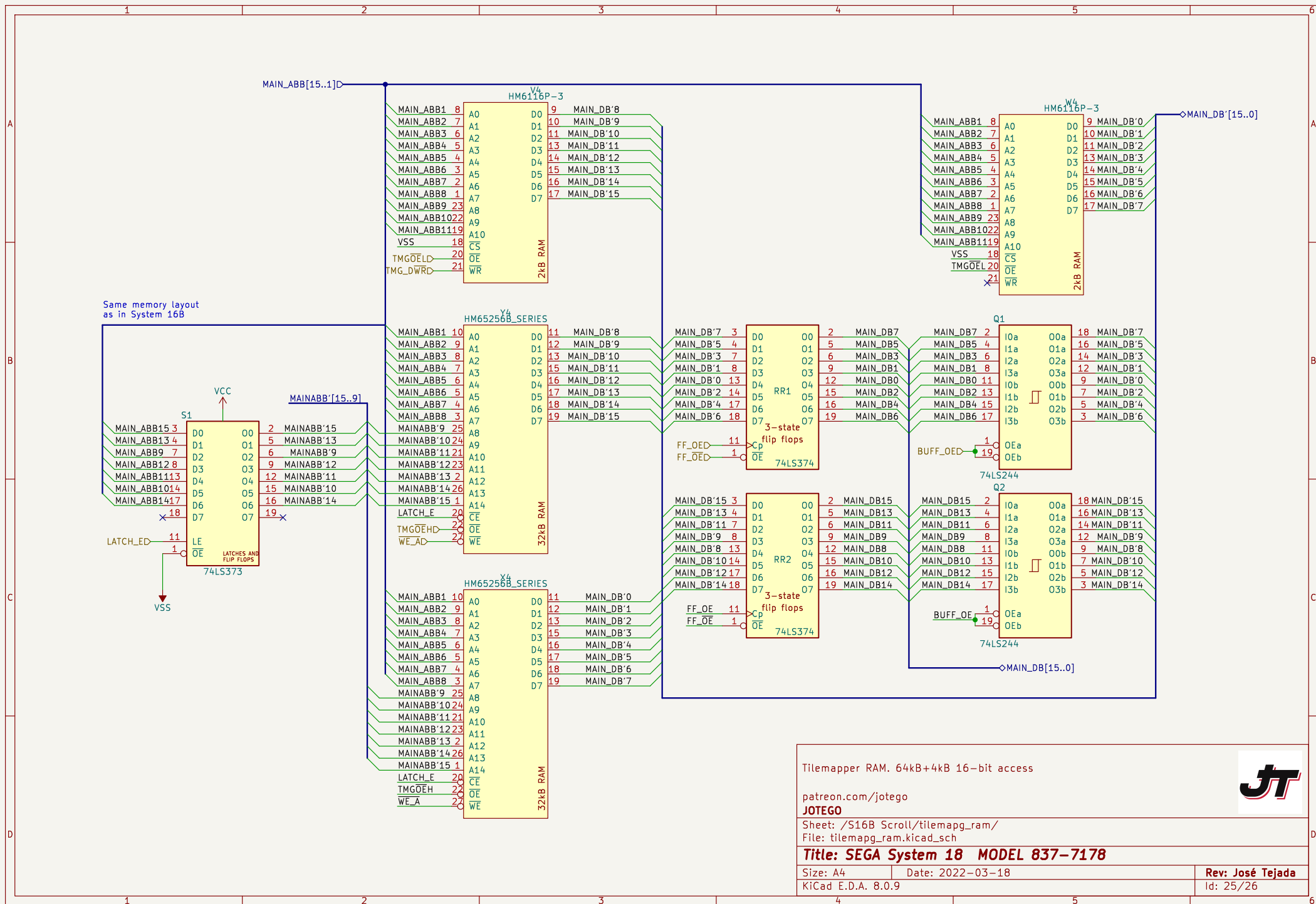
Sheet: /Sound/amplifier/audio_amplifier/
File: audio_amplifier.kicad_sch

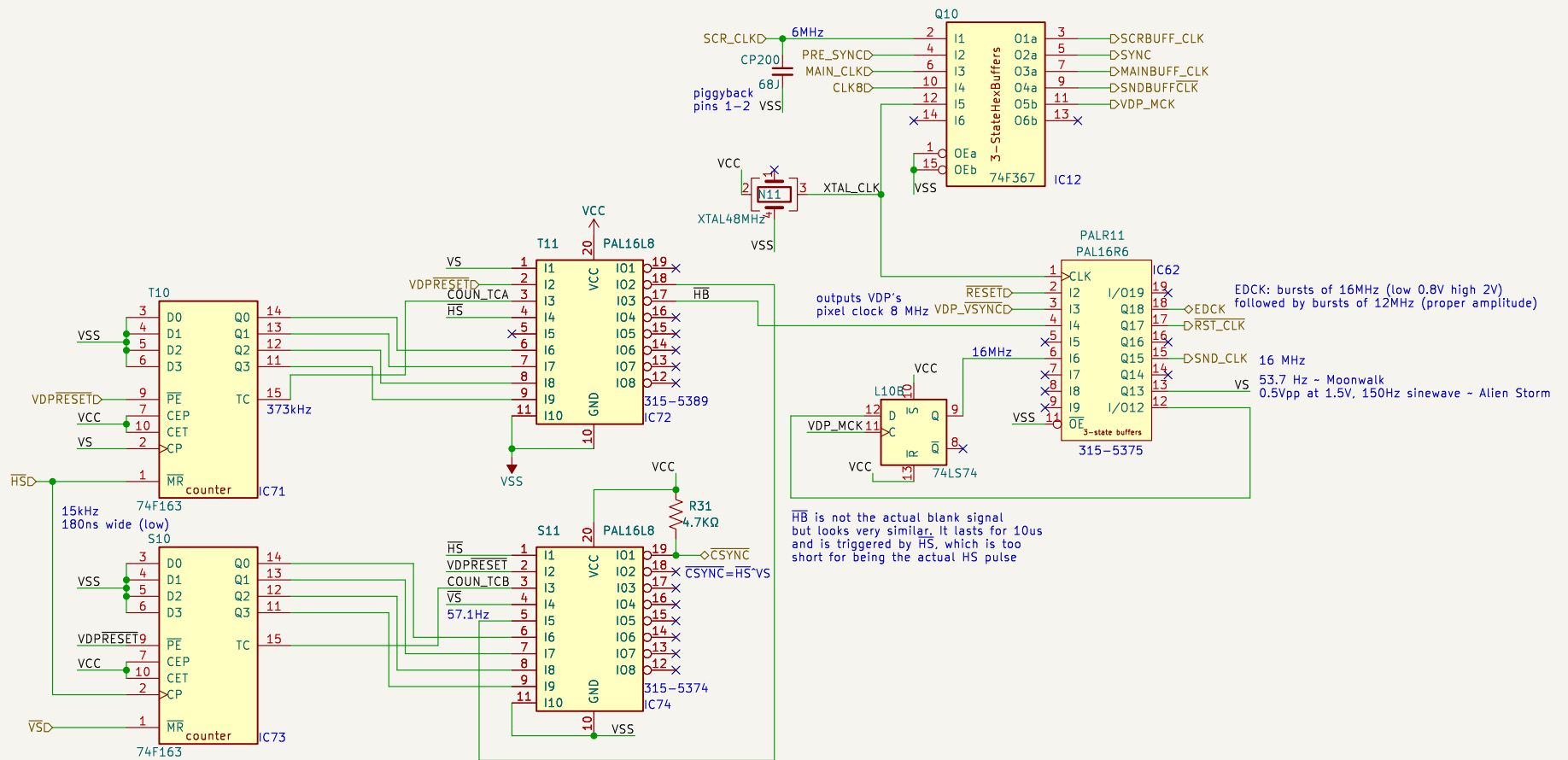
Title: SEGA System 18 MODEL 837-7178

Size: A4	Date: 2022-03-18	Rev: José Tejada
KiCad E.D.A. 8.0.9		Id: 23/26









Many of these signals do not look well formed. The PCB uses 74F series for many of these chips, but still cannot get proper digital waveforms.

Removing IC74 breaks the HS/VS alignment between the two video chipsets. The VDP video appears shifted on the screen.

But the HS/VS rate is not affected, so the VDP image still looks stable.

VDP pixel clock and csync generation
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Sheet: /Video Sync/
File: buffers.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4 Date: 2022-03-18

KiCad E.D.A. 8.0.9

Rev: José Tejada

Id: 26/26

